Recurrence relation
&
Generating Functions

There are some basic technique of counting, such as permutation, combination, etc. Many counting problems can not be solved by using the basic techniques of counting. Such problems can be solved by recurrence relation and generating function

A sequence is an ordered list of numbers that follow a specific pattern or rule. Each number in the sequence is called a term. A sequence is just a list of numbers in a specific order, with each number following a certain rule or pattern. Understanding sequences is fundamental in mathematics, as they appear in various contexts, from simple counting to complex calculations in calculus and number theory.

Arithmetic Sequence
Geometric Sequence
Fibonacci Sequence

A recurrence relation is an equation that defines a sequence based on previous terms in the sequence.

Simply, Recurrence Relation: A formula to find each term of a sequence based on previous terms. recurrence relation is like a recipe that tells you how to find the next number in a sequence if you know the ones that came before.

Recurrence relation

A recurrence relation is an equation that defines a sequence based on a rule that gives the next term as a function of the previous term(s).

The simplest form of a recurrence relation is the case where the next term depends only on the immediately previous term. If we denote the n th term in the sequence by x_n, such a recurrence relation is of the form $x_{n+1} = f(x_n)$ for some function f. One such example is $x_{n+1} = 2 - \frac{x_n}{2}$

Solving Recurrence Relations

•Basically, when solving such recurrence relations, we try to find solutions of the form $\mathbf{a_n = r^n}$, where r is a constant.

•$a_n = r^n$ is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2} + \ldots + c_ka_{n-k}$ if and only if

•$r^n = c_1r^{n-1} + c_2r^{n-2} + \ldots + c_kr^{n-k}$.

•Divide this equation by r^{n-k} and subtract the right-hand side from the left:

•$r^k - c_1r^{k-1} - c_2r^{k-2} - \ldots - c_{k-1}r - c_k = 0$

•This is called the **characteristic equation** of the recurrence relation.

Solving Recurrence Relations

- The solutions of this equation are called the **characteristic roots** of the recurrence relation.
- Let us consider linear homogeneous recurrence relations of **degree two**.
- **Theorem:** Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2.
- Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ if and only if $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$ for $n = 0, 1, 2, \ldots$, where α_1 and α_2 are constants.

- **See pp. 321 and 322 for the proof.**

Solving Recurrence Relations

- **Example:** What is the solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$ and $a_1 = 7$?

- **Solution:** The characteristic equation of the recurrence relation is $r^2 - r - 2 = 0$.
- Its roots are $r = 2$ and $r = -1$.
- Hence, the sequence $\{a_n\}$ is a solution to the recurrence relation if and only if:
- $a_n = \alpha_1 2^n + \alpha_2 (-1)^n$ for some constants α_1 and α_2.

Solving Recurrence Relations

- Given the equation $a_n = \alpha_1 2^n + \alpha_2(-1)^n$ and the initial conditions $a_0 = 2$ and $a_1 = 7$, it follows that
- $a_0 = 2 = \alpha_1 + \alpha_2$
- $a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1)$
- Solving these two equations gives us $\alpha_1 = 3$ and $\alpha_2 = -1$.
- Therefore, the solution to the recurrence relation and initial conditions is the sequence $\{a_n\}$ with
- $a_n = 3 \cdot 2^n - (-1)^n$.

Solving Recurrence Relations

- **Example:** Give an explicit formula for the Fibonacci numbers.
- **Solution:** The Fibonacci numbers satisfy the recurrence relation $f_n = f_{n-1} + f_{n-2}$ with initial conditions $f_0 = 0$ and $f_1 = 1$.
- The characteristic equation is $r^2 - r - 1 = 0$.
- Its roots are

$$r_1 = \frac{1+\sqrt{5}}{2}, \quad r_2 = \frac{1-\sqrt{5}}{2}$$

$$a_2 = -a_1$$

$$a_1 \cdot \frac{1+\sqrt{5}}{2} + (-a_1) \cdot \frac{1-\sqrt{5}}{2} = 1$$

$$a_1 \left(\frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} \right) = 1$$

$$a_1 \cdot \frac{(1+\sqrt{5}) - (1-\sqrt{5})}{2} = 1$$

$$a_1 \cdot \frac{2\sqrt{5}}{2} = 1 \Rightarrow a_1 \cdot \sqrt{5} = 1$$

$$a_1 = \frac{1}{\sqrt{5}}$$

$$a_2 = -a_1 = -\frac{1}{\sqrt{5}} \quad \boxed{a_1 = \frac{1}{\sqrt{5}}, \quad a_2 = -\frac{1}{\sqrt{5}}}$$

Solving Recurrence Relations

- Therefore, the Fibonacci numbers are given by

$$f_n = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$$

- for some constants α_1 and α_2.
- We can determine values for these constants so that the sequence meets the conditions $f_0 = 0$ and $f_1 = 1$:

$$f_0 = \alpha_1 + \alpha_2 = 0$$

$$f_1 = \alpha_1 \left(\frac{1+\sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1-\sqrt{5}}{2} \right) = 1$$

Solving Recurrence Relations

- The unique solution to this system of two equations and two variables is

$$\alpha_1 = \frac{1}{\sqrt{5}}\ ,\ \alpha_2 = -\frac{1}{\sqrt{5}}$$

- So finally we obtained an explicit formula for the Fibonacci numbers:

$$f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$